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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 02/28/2007 |
| **PI Name:** | Oman, Charles M. Ph.D. | | |
| **Project Title:** | Human Orientation and Sensory-Motor Coordination in Prolonged Weightlessness (ISS HRF Expt. E085) | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Behavior and performance | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 02139-4301 | **Congressional District:** | 7 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 96-OLMSA-01 |
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| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA JSC |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | Deselected. | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | McIntyre, Joseph ( College de France )<br>Berthoz, Alain ( College de France )<br>Liu, Andrew M. ( Massachusetts Institute of Technology )<br>Howard, Ian P. ( York University )<br>Young, Laurence R. ( Massachusetts Institute of Technology )<br>Beall, Andrew C. ( University of California Santa Barbara )<br>Smith, Theodore R. ( CNET Networks ) | | |
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| **Task Description:** | This International Space Station Utilization Flight Experiment (E085), is being conducted as part of Human Research Facility (HRF) Experiment: Visuomotor and Orientation Investigations in Long Duration Astronauts (VOILA). VOILA merges and extends into long duration flight two investigations originally developed for the 1998 Neurolab space shuttle mission. The science team includes investigators from the United States, Canada, France, and Italy. This experiment addresses the primary area of Space Physiology and Countermeasures. The Neurovestibular Adaptation sections of the NASA Critical Path Roadmap identift several risks of long-duration spaceflight requiring scientific investigation and countermeasure development. The information from our experiments addresses several critical questions identified within the highest ranked risk (#33), Disorientation and Inability to Perform Landing, Egress, or Other Physical Tasks, Especially During/After G-Level Changes. Our experiments will provide quantitative information useful for understanding the underlying mechanisms of the effects of weightlessness on disorientation, visual reorientation illusions (the downs), inversion illusions, EVA acrophobia, motion sickness, and human sensorimotor coordination (Critical Question 9.09) involved with reaching and catching, and the development of practical countermeasures (Critical Question 9.08) such as preflight visual orientation training. No scientific data on these phenomena is yet available from long duration space flights, and our experiment provides the opportunity to rapidly move from CRL0 to CRL5 in these areas. In terms of likelihood and consequences, spatial disorientation is considered the highest(red/one) neurovestibular risk. Our experiments also provide information relevant to the secondary area of human behavior and performance within ineffective habitat critical path risk (#20), as relates to spacecraft interior layout and work area design.<br>On earth, gravity provides a reference that influences how we recognize objects in our field of view and perceive their shape and orientation. It also influences our expectations of how objects will behave when thrown or dropped. The orientation of familiar visual objects and surfaces in turn influence our own perceived orientation. Gravity also helps the central nervous system align the various frames of reference used in movement control. All of these processes are fundamentally altered in weightlessness, as evidenced by the visual reorientation, inversion, and proprioceptive illusions frequently reported in-orbit by astronauts, and also by the surprisingly long lasting after effects seen, particularly after prolonged space flight. The experiment utilizes the HRF Rack 2 graphics workstation as a tool to study these processes during and after orbital flight. In VOILA, restrained and free-floating subjects wear a wide field of view, color stereo head mounted display, and view controlled visual scenes in five different test paradigms which assess: (1) the influence of scene symmetry, rotation, haptic cues, and expected orientation on static and dynamic self tilt (Virtual Tilting and Tumbling Room Tests)and visual reorientation illusions (VRIs); (2) the onset of x-axis illusory linear self-motion without haptic cues (Linear Vection Test); (3) the effect of perceived orientation on visual object recognition and shape recognition (Object Recognition Tests); (4) whether information used in grasping remembered objects is stored in head fixed, body fixed, or exocentric reference frames (Virtual Grasping Test); and (5) how the timing of catching movements depends on anticipation of downward acceleration (Virtual Catching Test). The 15-45 minute long test modules are based on existing 1-G paradigms, and can be selected and performed by an astronaut in an automated fashion using Session Manager software. Two preflight, three inflight, and three postflight performances of each test are planned on each ISS increment. An engineering team from the MIT Center for Space Research is designing the suite of virtual reality hardware and software for the workstation, including a head mounted display, hybrid inertial-optical 6 degree-of-freedom head tracker, a subject input device, and a subject restraint system. The latter consists of a folding chair-like body restraint, and a pole equipped with a swivel joint for free floating experiments. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The vertebrate nervous system evolved in an environment where the stimulus to the body's multiple gravireceptive senses invariably changes whenever the orientation of the body is altered. Conducting experiments in orbital flight removes these static gravireceptor cues and helps us understand the role of gravity in fundamental sensory, motor, and cognitive processes subserving spatial orientation and movement control in daily life on earth. We usually only become aware of these functions when they are compromised by inner ear or central nervous system disease as our everyday lives are then profoundly affected. More than 90 million Americans suffer from some type of balance disorder which can cause difficulty walking at night, impaired vision (particularly when moving), and injurious falls, or even incapacitating bouts of vertigo and nausea. Humans with hippocampal lesions or Alzheimer's disease show impairments on a wide variety of spatial and navigational tasks.<br>Insight into how astronauts adapt to microgravity should provide valuable information into the normal functioning of the human nervous system. Strategies used by human subjects to orient and navigate in three dimensions and compensate for the lack of gravity may also suggest how the neural disorders mentioned above might be treated. There is also currently considerable interest in the development of new methods for evaluating a patient's ability to use visual and proprioceptive cues to maintain balance and orientation, and for improving motor control function via rehabilitative training. Portable head mounted displays and head and hand tracking systems, akin to those used in this experiment, may prove useful for diagnostic testing and rehabilitative training. |
| **Task Progress:** | No progress report received this reporting period. |
| **Bibliography Type:** | Description: (Last Updated: 12/18/2024) |